

Product Manual

1. Product Name

HTP-300Y Submersible Pressure Transmitter

2. Product Description

The HTP-300Y is a submersible pressure transmitter designed for liquid level and pressure measurement in marine, industrial, and water management applications. Featuring a stainless steel sensing diaphragm and IP68 protection rating, it is suitable for long-term immersion in liquids. The device provides stable and accurate output signals for integration with control systems, PLCs, and monitoring equipment.

3. Specifications

Power supply: DC24V default, optional 9 to 24VDC, 3.1 to 5V

Rated voltage: 24VDC nominal

Power options: 9 to 24VDC or 3.1 to 5VDC depending on output type

Material: 316L stainless steel sensing diaphragm, aluminum alloy housing, polyurethane cable

Size: [To be filled by seller]

Cable length: 1.5 m default, optional 10 m, 15 m, 20 m or customized

Protection level: IP68

Model: HTP-300Y

Measurement range: -100 kPa to 100 MPa customizable within -0.1 to 100 MPa

Pressure type: Gauge, Absolute, Sealed

Output signal: 4 to 20 mA, 0 to 5 V, 0 to 10 V, RS485

Accuracy: ± 0.5 percent FS typical, optional ± 0.2 percent FS, ± 0.1 percent FS

Compensated temperature range: -10 to 70°C

Operating temperature: -20 to 80°C

Zero temperature drift: ± 0.5 percent FS maximum

Sensitivity temperature drift: ± 0.5 percent FS maximum

Overpressure: 2 times FS for ranges up to 10 MPa, 1.5 times FS for ranges above 25 MPa

Vibration resistance: 20G at 20 to 5000 Hz

Sampling rate: 2400 samples per second

Response time: 10 ms

Insulation resistance: 100 M Ω at 250 VDC

Pressure units supported: MPa, kPa, bar, psi, kgf per cm²

Package contents: Pressure transmitter with integrated cable

4. Instructions for Use

Preparation

Verify that the selected pressure range, output signal type, and power supply match the system requirements. Confirm that the supply voltage is within the specified range. Ensure that the installation environment is within the permitted temperature limits.

Installation

Mount the transmitter vertically and immerse the sensing probe fully into the liquid to the required depth. Avoid contact between the sensing diaphragm and hard surfaces. Ensure that the cable is securely fixed to prevent mechanical stress on the probe.

Electrical Connection

Connect the power supply and output wires according to the wiring definition provided by the seller. For 4 to 20 mA output, connect in series with the load and ensure correct polarity. For voltage or RS485 output, connect to the corresponding input terminals of the control system.

Operation and Data Reading

After power is applied, allow a short stabilization time. The output signal corresponds linearly to the selected pressure range. Read the measured value from the connected controller, display unit, or PLC according to its configuration.

5. Precautions

Ensure the supply voltage does not exceed the specified limits to prevent permanent damage.

Do not exceed the rated overpressure limits.

Avoid mechanical impact or deformation of the sensing diaphragm.

Use only in media compatible with 316L stainless steel.

Do not operate outside the specified temperature range.

When used in corrosive or conductive liquids, verify material compatibility before installation.

Disconnect power before performing wiring or maintenance.

6. Storage and Maintenance

Store the transmitter in a dry and ventilated environment within -20 to 80°C.

Avoid exposure to corrosive gases or excessive humidity during storage.

Clean the exterior with a soft cloth. Do not use abrasive or corrosive cleaning agents.

Periodically inspect the cable and sealing parts for damage to maintain IP68 protection.

7. Warning Information (EU Market Reference)

WARNING!

Risk of electric shock. Ensure power is disconnected before installation or maintenance.

Do not use the device in explosive atmospheres unless specifically certified.

Improper installation may result in inaccurate measurement or equipment damage.

Keep out of reach of children.

Proper disposal: Follow WEEE directives for electronic waste.

Note: The above warning information is for reference only. Final warning language must comply with the official language requirements of the destination EU country.

8. Compliance Disclaimer

This document is provided for product instruction purposes only.

It does not constitute legal advice.

The seller is responsible for ensuring compliance with destination market regulations.